Elementary Graph Algorithms

(BFS)

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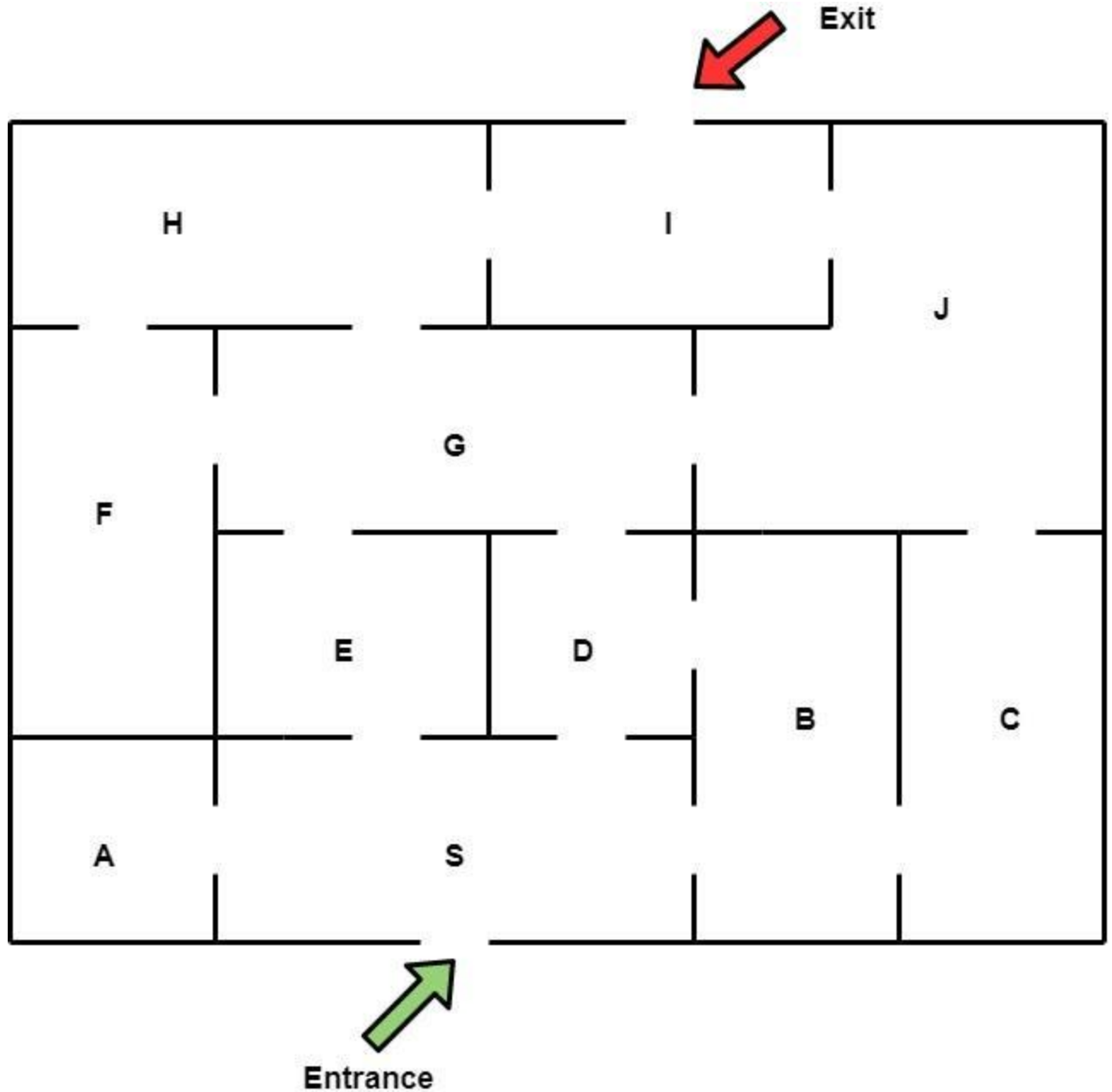
Exit
H
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Entrance

BREADTH-FIRST SEARCH

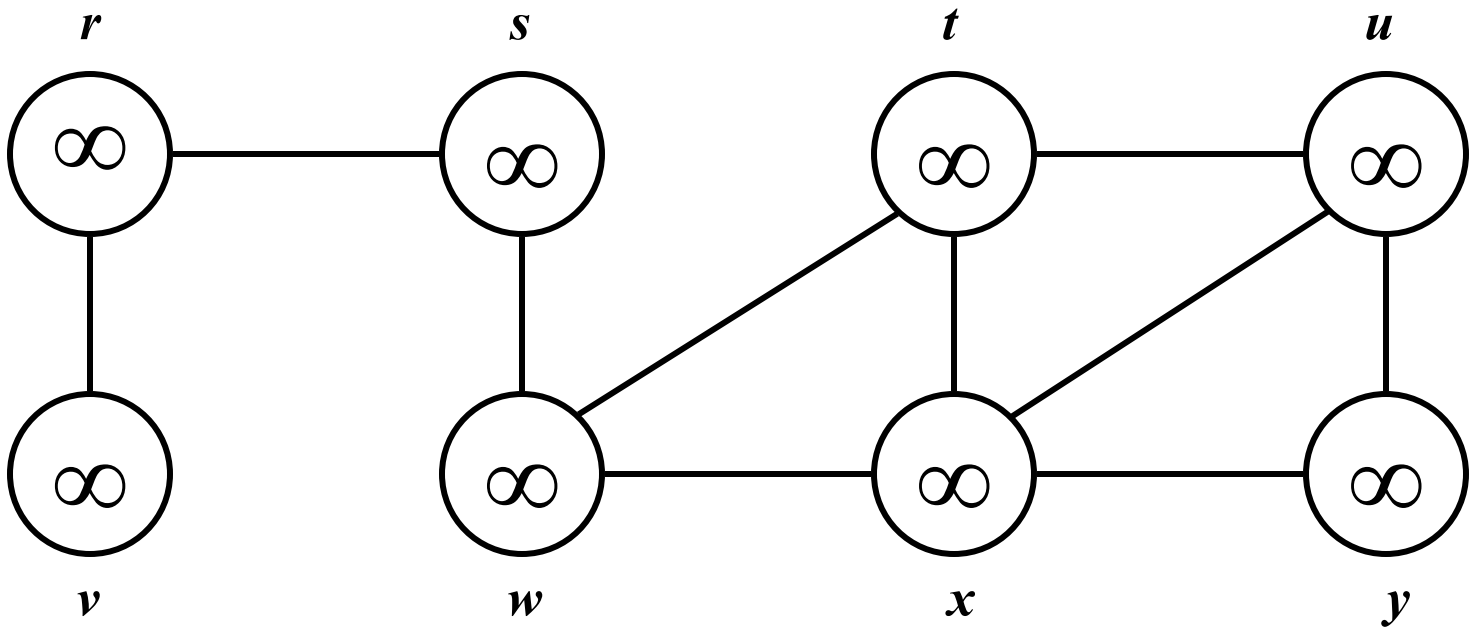
❖ Breadth-First Search (BFS)

- ✓ Scan a graph, and turn it into a 'Breadth-first (spanning) tree'
  - Tree with discovered vertices* and explored* edges.
  - One vertex at a time
  - Pick a source vertex, s to be the root.
  - Find ("discover") its children, then their children, etc.
  - For any vertex reachable from s,
    the path in the tree from s to v corresponds to a "**shortest path**' from s to v in G.
- ✓ * 'Discover' the vertex : The vertex is first encountered.
- ✓ * 'Explore' the edge : The edge is first examined.

BREADTH-FIRST SEARCH

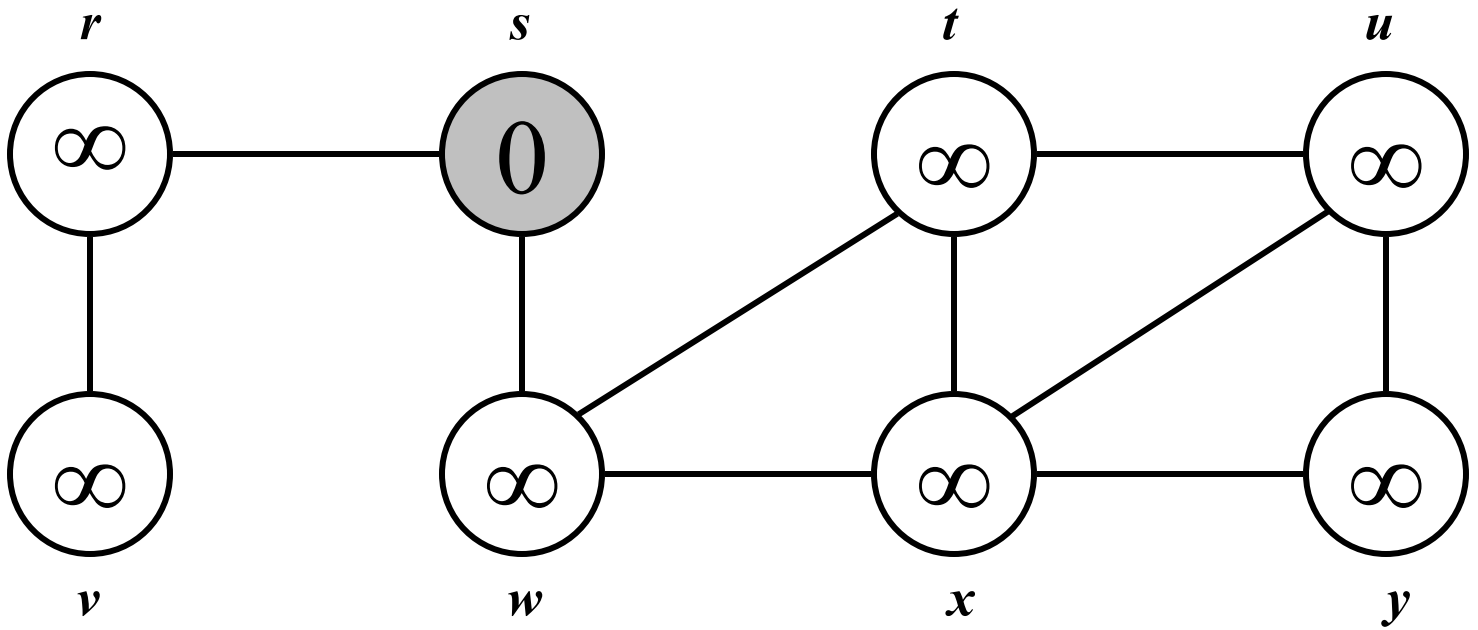
- **Again we will associate vertex "colors" to guide the algorithm.**
  - ✓ White vertices have not been discovered.
    - All vertices start out white.
  - ✓ Gray vertices are discovered but not finished.
    - They may be adjacent to white vertices.
    - The vertex is added to breadth-first tree.
  - ✓ **Black** vertices are discovered and finished.
    - They are adjacent only to **black** and gray vertices.
- **As soon as the white vertex is discovered**
  - ✓ Turn its color to 'gray'.
  - ✓ Discover all of its white neighbor.
  - ✓ After all of its neighbors are discovered, turn its color to **black** (means it is finished).
- **Discover vertices by scanning adjacency list of gray vertices.**

BREADTH-FIRST SEARCH: EXAMPLE



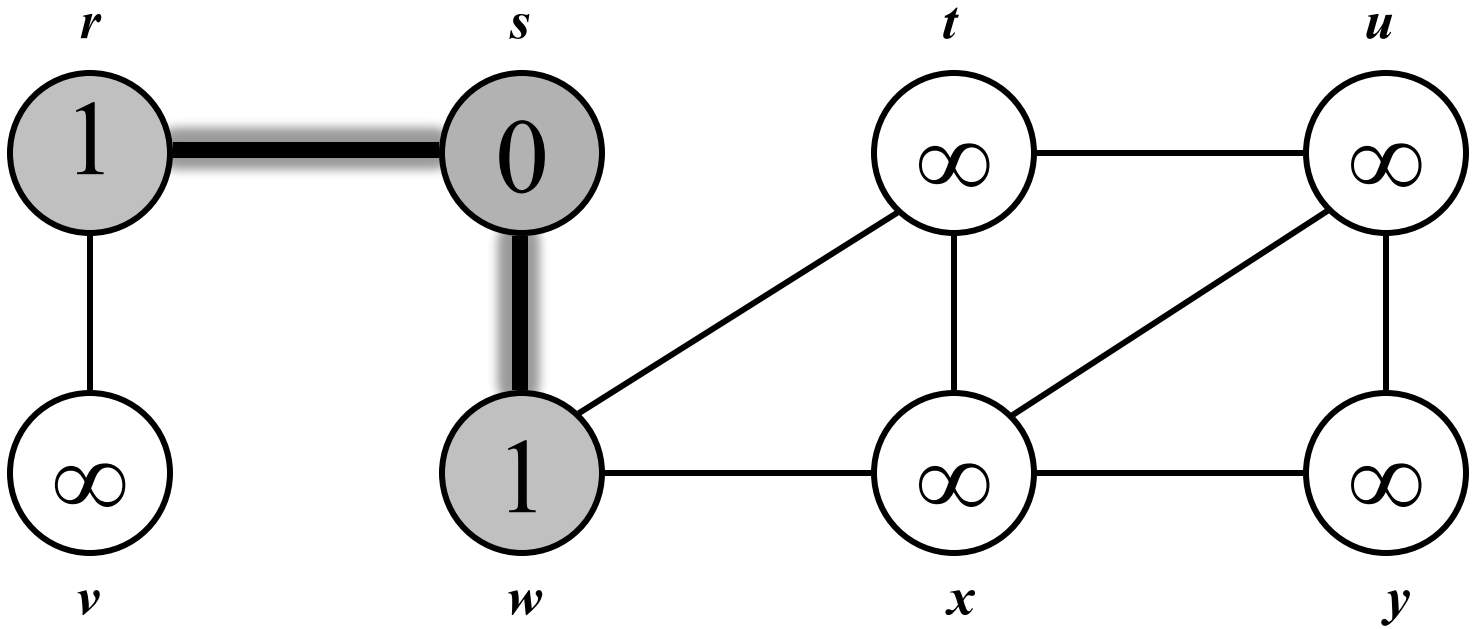
Vertex is an island and edge is a bridge. BFS is visualized as many simultaneous (or nearly simultaneous) explorations from the starting point and spreading out independently.

BREADTH-FIRST SEARCH: EXAMPLE



Q: | s |

BREADTH-FIRST SEARCH: EXAMPLE



Q: | w | r |

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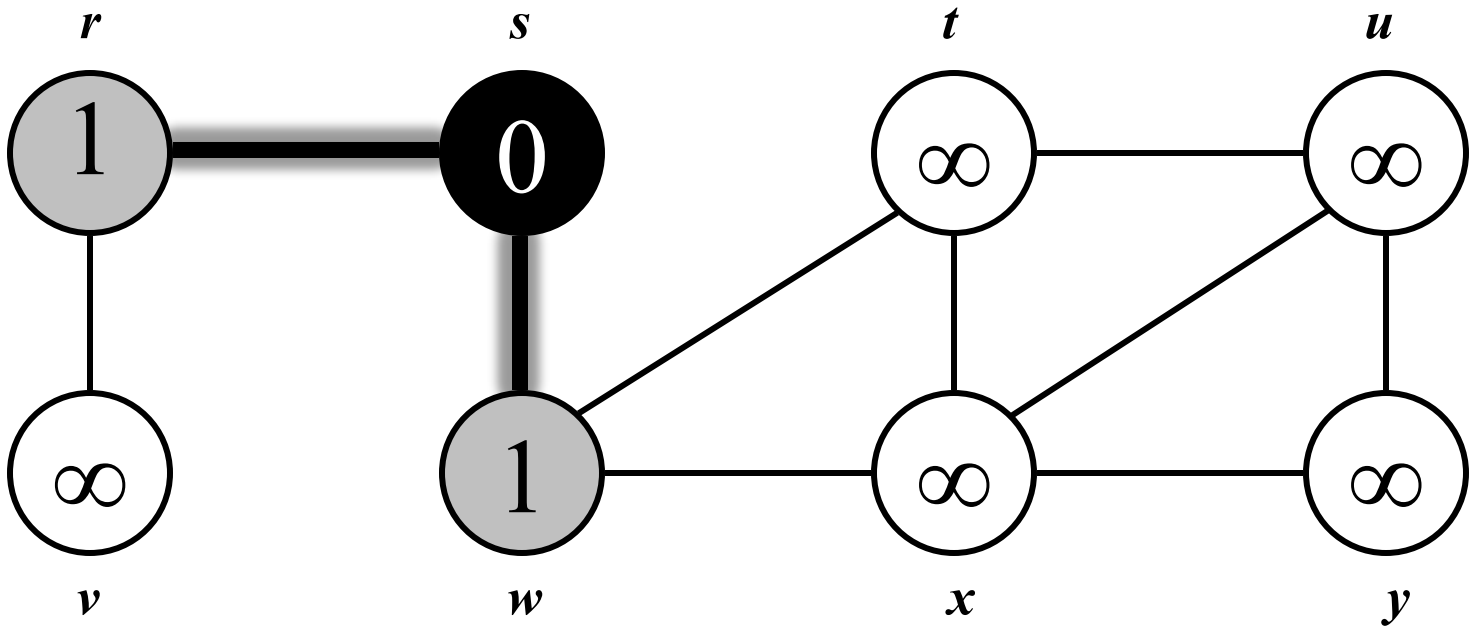
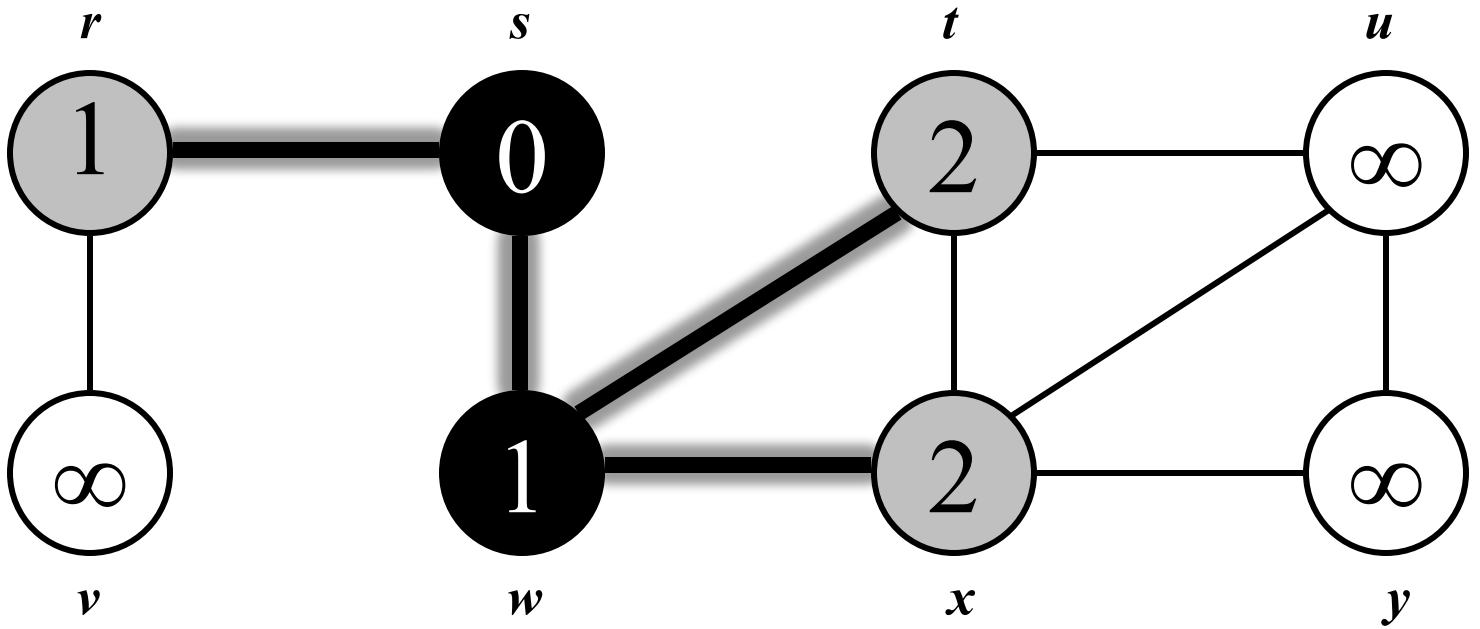


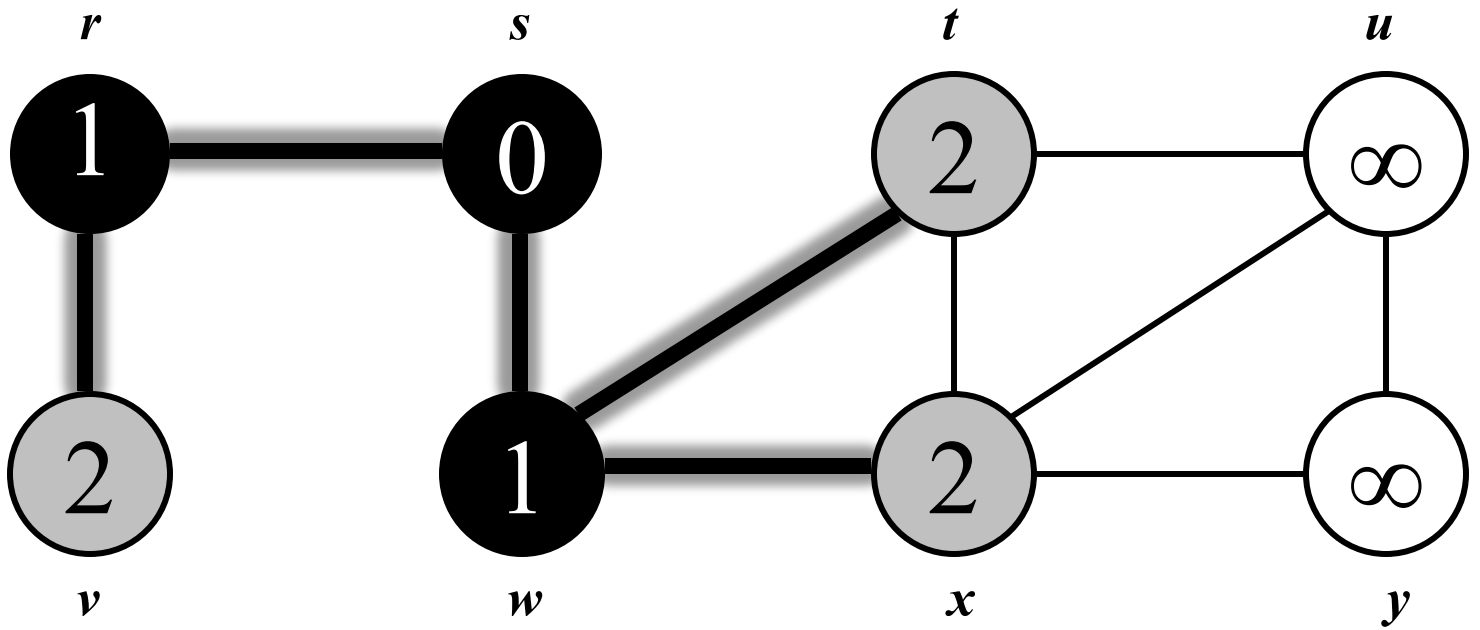
Q: | w | r |

BREADTH-FIRST SEARCH: EXAMPLE



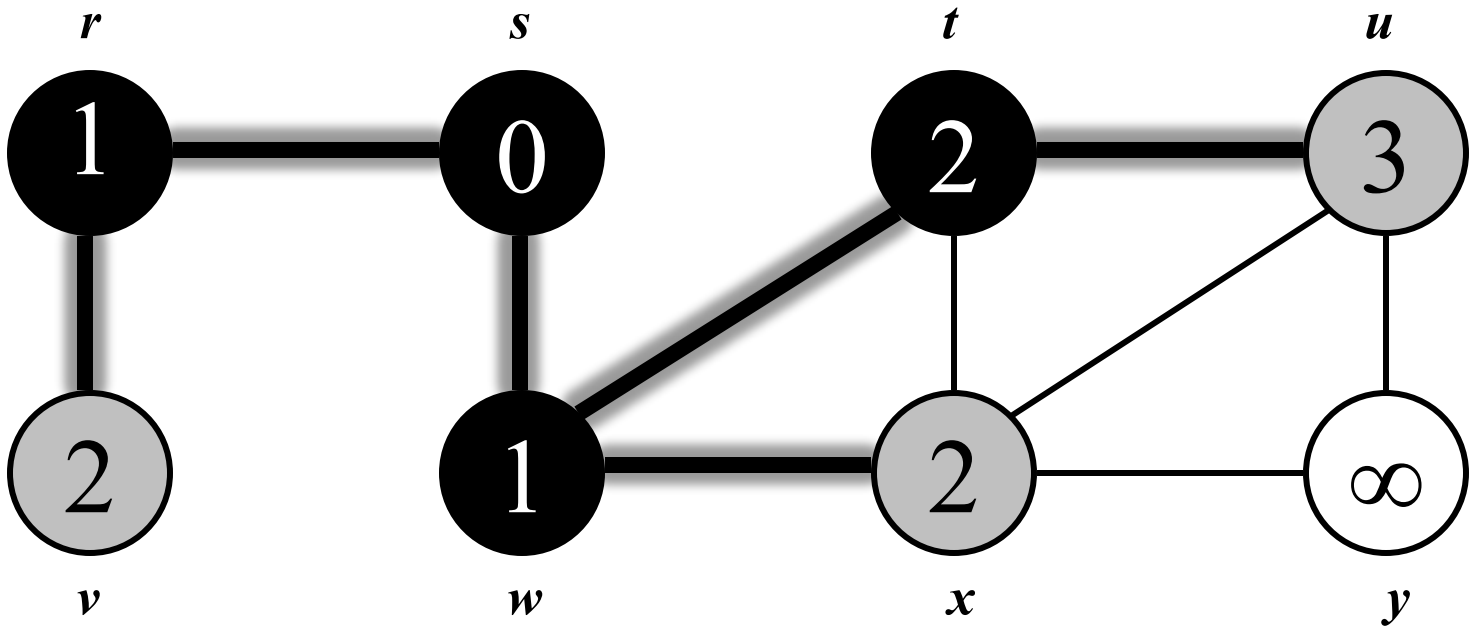
Q: | r | t | x |

BREADTH-FIRST SEARCH: EXAMPLE



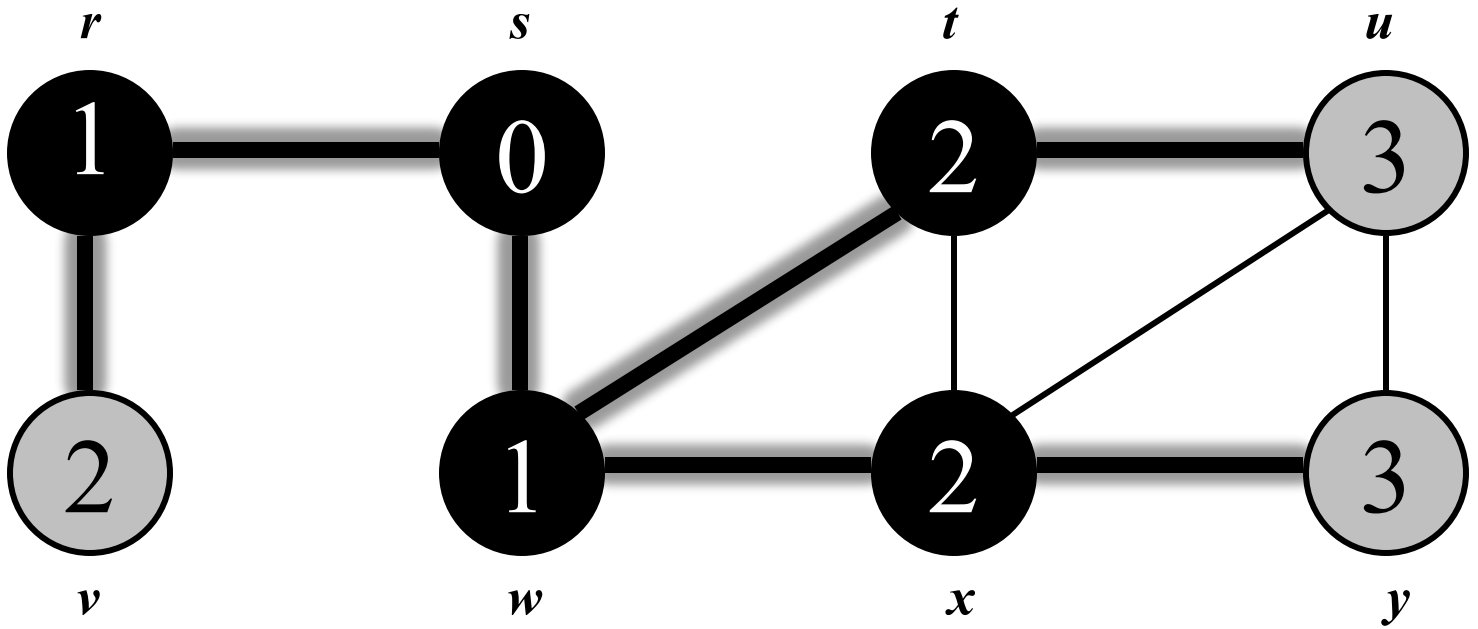
Q: | t | x | v |

BREADTH-FIRST SEARCH: EXAMPLE



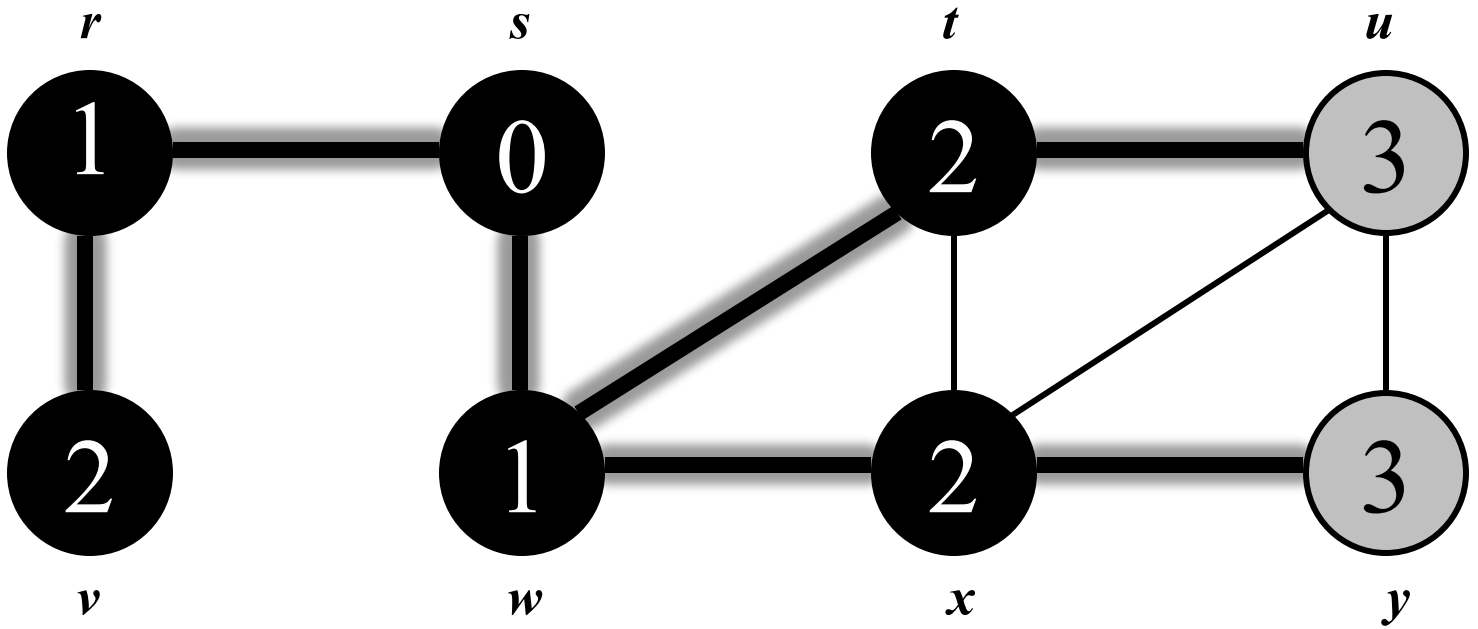
Q: | x | v | u |

BREADTH-FIRST SEARCH: EXAMPLE



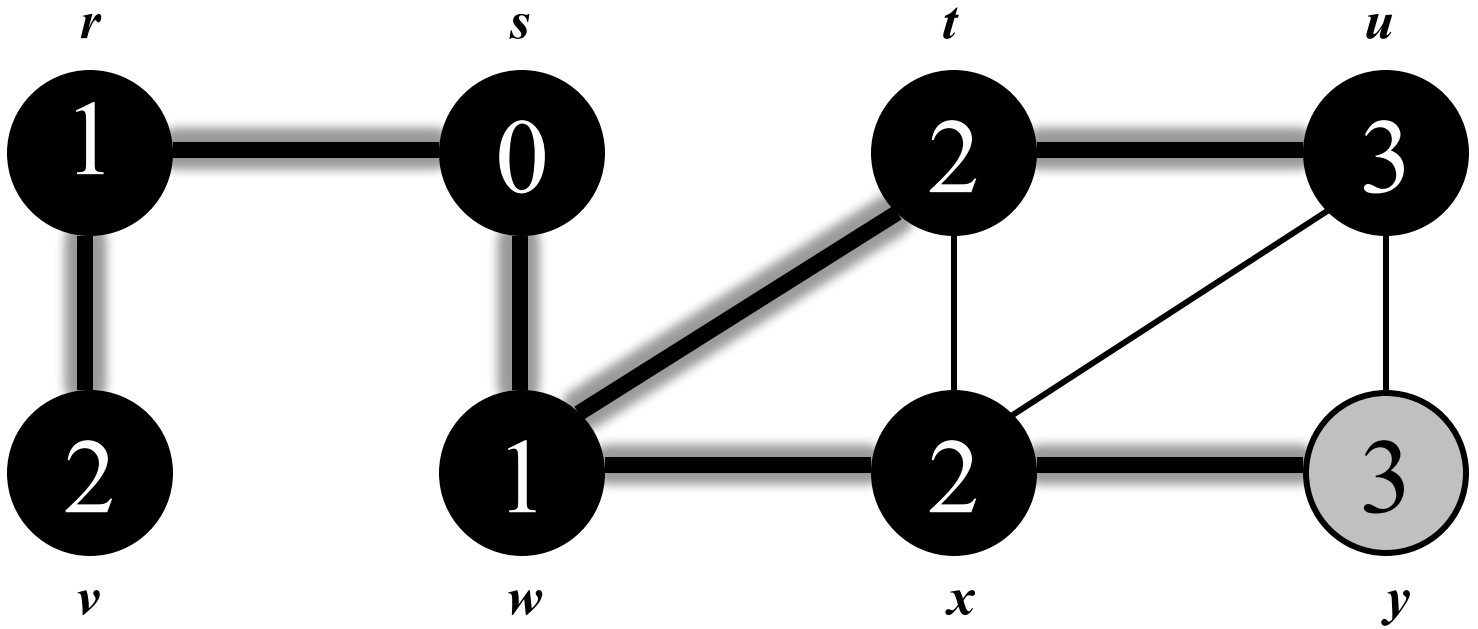
Q: | v | u | y |

BREADTH-FIRST SEARCH: EXAMPLE



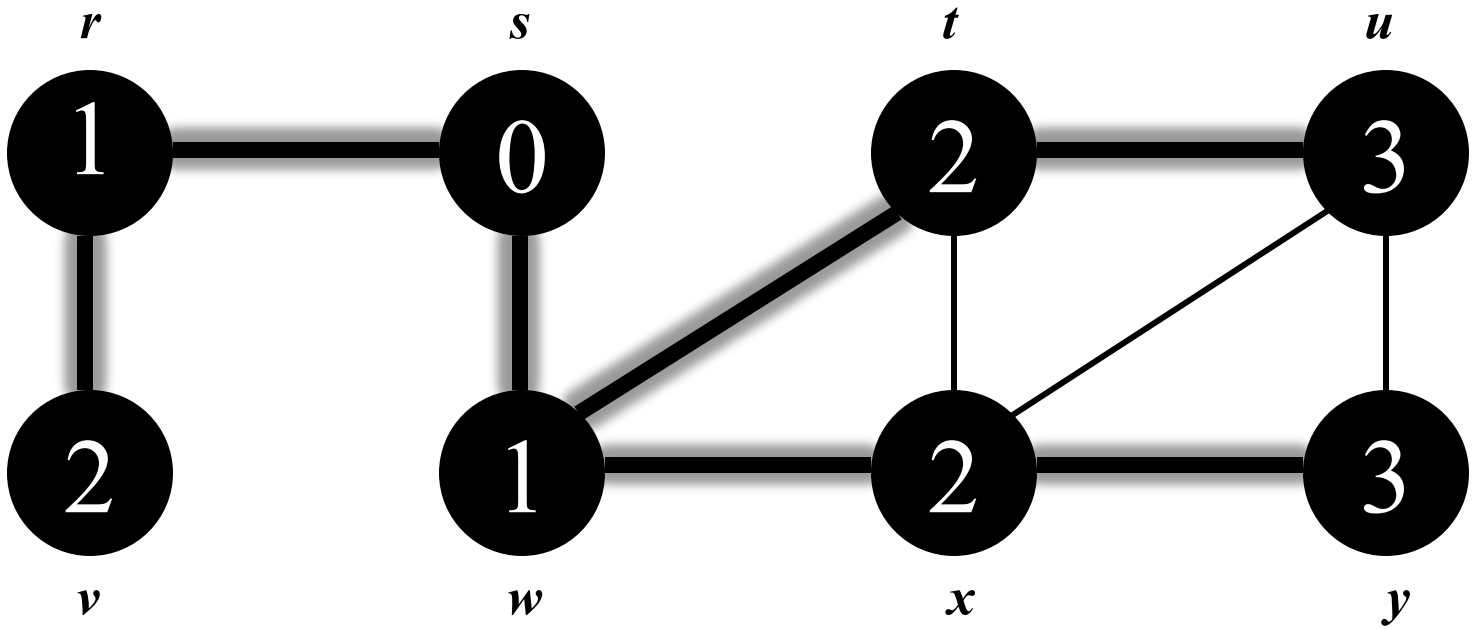
Q: | u | y |

BREADTH-FIRST SEARCH: EXAMPLE



Q: | y |

BREADTH-FIRST SEARCH: EXAMPLE



breadth-first tree :

Q: $\emptyset$ consists of vertices and explored edges

BREADTH-FIRST SEARCH

```
BFS(G, s)                                   // s: source vretex
  for each vertex u ∈ V – {s}
     do d[u] = ∞;   π[u] = NIL;             // π : predecessor (parent in the tree)
  color[s] = GRAY;  d[s] = 0; π[s] = NIL;
  Q = ∅
  ENQUEUE(Q, s)
  while Q ≠ ∅
    do u = DEQUEUE(Q)
       for each v  ∈ Adj[u]
           do if color[v] = WHITE
                  then color[v] = GRAY
                       d[v] = d[u] + 1
                       π[v] = u
                       ENQUEUE(Q, v)
     color[u] = BLACK
```

ANALYSIS

Running time of Breadth-First search:

A. Each node is enqueued once. (white → gray) : $\Theta(V)$

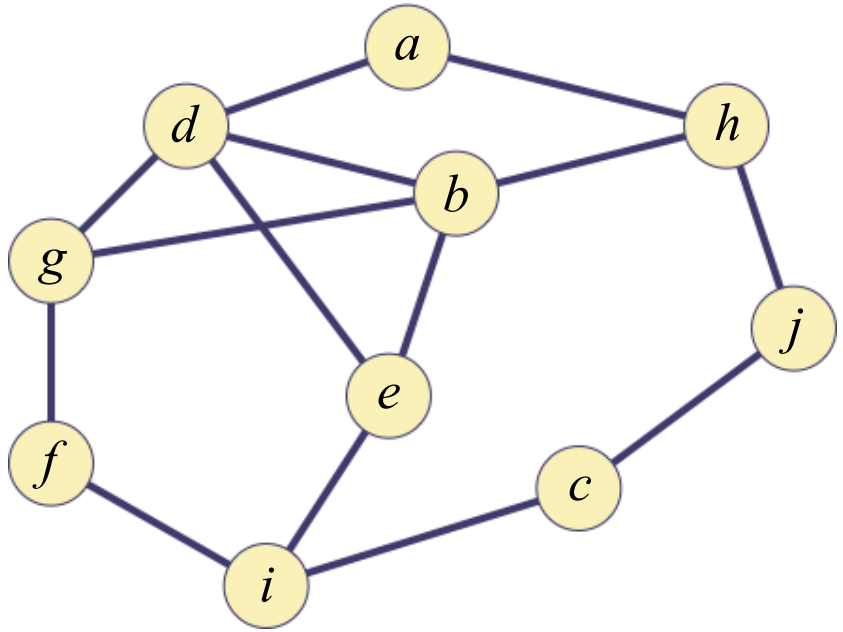
B. Each node is dequeued once. (gray → black) : $\Theta(V)$

C. Each adjacency list is scanned only once. : $\Theta(E)$

Overall running time is $\Theta(V + E)$.

EXERCISE

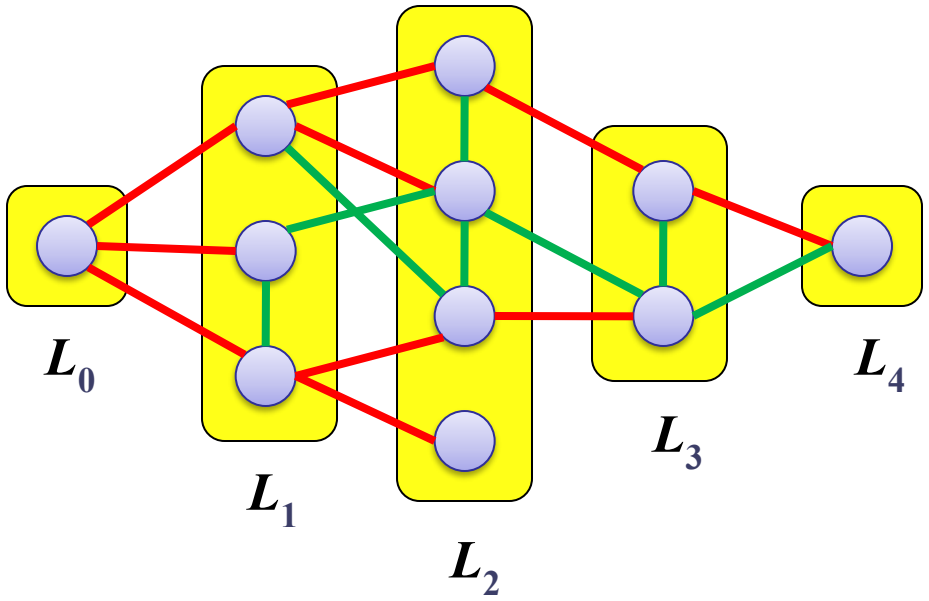
- **For the following graph,**
  - ✓ Build breadth-first (spanning) tree with vertex 'a' as root. Assume the vertices are in alphabetical order in the Adj array and that each adjacency list is in alphabetical order.
  - ✓ What is the minimum distance of vertex c from the root?



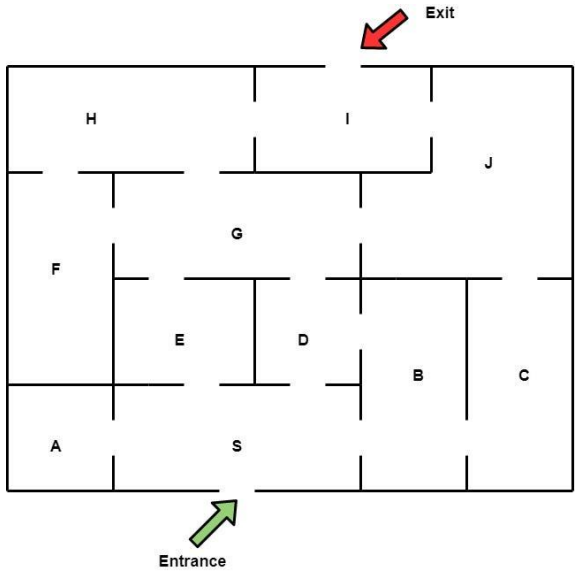
PROPERTIES OF BREADTH-FIRST SEARCH

Theorem: Breadth-first search visits the vertices of G by increasing distance from the root. BFS builds breadth-first tree, in which paths to root represent shortest paths in G.

Theorem: *For every edge* (v, w) *in* G *such that* $v \in L_i$ *and* $w \in L_j$, $|i - j| \leq 1$.



MAZE SOLVING USING BREADTH-FIRST SEARCH

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Thank you!